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A Comprehensive Approach to Gold Price Prediction Using Machine Learning and Time Series Models

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***Abstract*—This paper provides a comprehensive analysis of gold rate fluctuations in the USA and introduces an innovative method for more accurately predicting gold price changes using advanced machine-learning techniques. The dataset consists of daily gold rate data from January 1, 1985, to September 8, 2023, expressed in this country's national currency. However, for our research, we specifically utilized the data expressed in United States Dollars (USD). Several machine learning models, ARIMA, LSTM, FB Prophet, SVM, Random Forest, and XGBoost, were employed to model and predict these time series data. The predictive performance of these methods is compared, providing insights into their effectiveness in capturing non-linear and long-term trends in gold prices. The outcomes reveal how machine learning could guide financial time series forecasting better and bring to light the most powerful clues about the way gold prices are moving on the global stage.**

***Keywords*—Gold Price Prediction, ARIMA, SVM, Random Forest, XGBoost , LSTM.**

I. Introduction

Savings and investments are essential aspects and play a crucial role in everyone's life. Investing means putting current funds to work to gain a profitable return in the future. There are many investment options to choose from, among which, the most common are stocks, deposits, commodities, and real estate, each one of them possessing their particular risk and reward profile [1]. Gold, in particular, is seen by many investors as a desirable investment due to its rising value and wide range of applications. It is often set as "last option to save financial assets", which indicate that investors seek gold when world capital markets in the developed world fail to deliver the expected returns [2]. Conversely, gold has always been essential to the world economy. Conversely, gold has always been essential to the world economy, Due to its intrinsic value and inverse correlation with currency devaluation, gold prices have been used as an indicator of economic stability and investor confidence across different regions [3].

Over the last few decades machine learning techniques gained popularity for forecasting time series data because they can handle complex, nonlinear patterns. [4], [5]. Traditional methods like ARIMA are efficacious in short-term forecasting [6], [7]. Nevertheless, some incredibly superior models, like (LSTM) networks, may be artifacts of long-term trends [8], [9]. Recent approaches, like Random Forest, XGBoost, and SVM, have also shown strong results in financial forecasting This study by its nature is a bit tricky, which is aimed at finding out if the use of machine learning models are on par with or superior than other means in case of the gold price forecasting in the USA [10]. In this study, historical price data has been used to evaluate predictive models, comparing their performance in terms of accuracy and reliability. By conducting this comparative analysis, this paper aims to provide insights that could help in decision-making for investors and policymakers.

II. Literature Review

For centuries, gold has been seen as a dependable way to invest an asset due to its intrinsic value and historical role as a store of wealth. Numerous studies highlight its ability to serve as a hedge against inflation and currency devaluation. Gold is referred to as a "protective asset" by Baur and Lucey (2010), particularly during periods of market volatility and economic uncertainty [11]. According to research by Beckmann et al. (2015), gold prices frequently fluctuate in the opposite direction of stock markets, highlighting its value as a hedge against unstable financial markets [12]. As the global economy becomes more interconnected, the demand for gold has expanded across regions, with price fluctuations driven by geopolitical events, inflationary pressures, and changes in supply and demand. Studies, like that of Shafiee and Topal (2010), emphasize the importance of historical trends and macroeconomic indicators in determining gold prices [13]. Time Series Forecasting in Financial Markets, including gold prices, is difficult because of the unpredictable and intricate nature of market dynamics. For centuries the ARIMA model and, on the whole, trade journals, have been one of the main applications for Financial Forecasting. While effective for short-term predictions, these models struggle as it helps identify long-term patterns and complex connections in time series data [14]. The rise of machine learning techniques offers new avenues for improving the accuracy of financial time series predictions. Hyndman and Athanasopoulos (2018) note that machine learning models can handle large datasets and identify patterns missed by classical statistical models [15]. Advanced algorithms like LSTM, Random Forest, and XGBoost have shown promise in capturing complex trends in time series data.

Financial markets' rising interest in machine learning is owing to its feature of handling non-linear relationships in the times series data, which it models using machine learning has resulted in the increase of machine learning in financial markets. Traditional models like ARIMA and GARCH are effective for linear dependencies but often fall short in handling the volatility and shifts that characterize financial datasets [16] . The study by Patel et al. (2015) showed that machine learning methods such as RF and SVM turned out to be the most accurate and efficient in stock price prediction, finding that ensemble models like Random Forest offered better accuracy than traditional methods [17]. Similarly, Nath and Behara (2020) showed that LSTM networks are a great fit for capturing long-term patterns. price prediction, Identifying sequential relationships in time series data [18].

A. LSTM Networks

A variant of RNNs, known as LSTM networks, have become more and more popular as they can store information over a long sequence, which is especially useful in time series forecasting. Gers et al. (1999) were the first to solve the vanishing gradient issue in RNNs using LSTMs, thereby letting the model keep long-term dependencies in mind [19]. In gold price forecasting, LSTMs have been used to not just simply model volatility but to record both transient and permanent patterns .

B. XGBoost and Random Forest

Ensemble methods like Random Forest and XGBoost are known for their robustness and ability to reduce overfitting [20]. Both models use multiple decision trees to enhance prediction accuracy. Chen and Guestrin (2016) highlighted XGBoost's popularity for its efficiency and accuracy in handling large datasets with complex features [21]. Random Forest has also demonstrated success in capturing non-linear dependencies in asset prices [22].

C. FB Prophet

FB Prophet, developed by Facebook, models time series data with daily and seasonal patterns [23]. It has been used successfully in financial domains, including stock price prediction. The Prophet model incorporates yearly, weekly, and daily seasonalities, making it suitable for time series with strong periodic components [24].

D. Challenges and Future Directions

Despite a number of developments, the difficulty of predicting financial time series is still high because markets are regularly changing and several external events continuously are taking place. While machine learning models show promise, their reliance on historical data can be a limitation when dealing with unprecedented events or market disruptions. Future research could be directed to modeling that consists of traditional statistical methods and machine learning algorithms. In addition, data collected from social media, news articles, and macroeconomic indicators are examples of alternative sources can be added to the predictive power. Zhang et al. (2021) suggest that multi-modal approaches leveraging diverse datasets could offer more accurate and robust forecasts for financial markets [25].

III. METHODOLOGY

The overall methodological approach is outlined in this section, Figure. 1 illustrates this process.

A. Data Collection and Preprocessing

The dataset utilized in this study comprises daily gold prices in United States Dollars (USD) covering the period from January 1, 1985, to September 8, 2023, offering valuable insights into long-term trends and market dynamics. This research specifically focuses on the behavior of gold prices within the context of the U.S. market, with all analysis conducted in USD, providing a detailed examination of gold price movements over nearly four decades. The data was preprocessed in several stages:

- **Handling Missing Values:** Missing values were addressed using interpolation and forward/backward filling methods, ensuring no gaps in the time series.
- **Normalization:** To enhance the effectiveness of machine learning models, the data was scaled using either MinMaxScaler or StandardScaler, which normalized the price values.
- **Dividing:** Eighty percent of the dataset was set aside for testing, and the remaining twenty percent was allocated for training.
- **Splitting the Dataset:** The dataset was split 80-20, with 80% of the data designated for training and 20% for testing.

B. Modeling Techniques

A variety of machine learning models have been applied to forecast gold prices. Below are the details of each model and the approach adopted in this study:

1) ARIMA: Among them is the ARIMA equation, which is created by fusing the autoregressive AR model with the moving average MA model and differencing to provide a stationary form for the data. In ARIMA(p, d, q), the ARIMA modal is expressed as;

$$\begin{aligned} Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} \\ + \theta_1 e_{t-1} + \theta_2 e_{t-2} + \cdots + \theta_q e_{t-q} + e_t \end{aligned} \tag{1}$$

Where:

- YtYtYt is the value at time t.
- ccc is a constant,
- ϕ are the AR coefficients (autoregressive coefficients),
- θ are the MA coefficients (moving average coefficients),

In this study, ARIMA (3, 1, 3) was applied to capture short-term trends. To get the accuracy of the model, MAE and MAPE were used.

2) FB Prophet: FB Prophet models time series through a decomposable structure, breaking it down into trend, seasonality, and holiday components.

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t \tag{2}$$

Where: The trend component is represented by g(t), the seasonal component by s(t), the holiday component by h(t), and the error term by ϵt, which takes into account any noise or random fluctuations in the data.

FB Prophet was applied to capture seasonal and daily patterns in gold prices. The model was evaluated based on MAE, MAPE, and accuracy metrics, and its forecasts were visualized using time-series plots.

3) SVM: The following optimization problem is addressed by Support Vector Machines (SVM) for regression, also known as Support Vector Regression (SVR):

$$\min_{\omega,b} \quad \frac{1}{2}\|\omega\|^2 \text{subject to} |y_i - (\omega^T \varphi(x_i))| \leq \epsilon \tag{3}$$

Where: ω represents the weight vector, b denotes the bias term, ϵ denotes the tolerance margin, xi denotes the input characteristics, and Yi denotes the target values. An RBF kernel was applied, and the model's parameters were optimized using grid search. SVM was evaluated using MAE, MAPE, and accuracy.

4) Random Forest: Random Forest regression builds multiple decision trees and averages their predictions. The prediction for a sample x_i is:

$$\hat{f}(x_i) = \frac{1}{N} \sum_{j=1}^{N} T_j(x_i) \tag{4}$$

Where: In this case, N is the number of trees in the model, and $T_j(x_i)$ is the forecast of the j-th tree for sample x_i. By modifying hyperparameters like the number of trees and the depth of each tree, the Random Forest model was improved. The model's effectiveness was measured using MAE, MAPE, and accuracy, with visualizations created to compare the actual and predicted prices.

5) XGBoost (Extreme Gradient Boosting): XGBoost uses a boosting framework in which each individual tree's prediction adds up to the total prediction.

$$\hat{y}_i = \sum_{k=1}^{N} f_k(x_i), \quad f_k \in F \tag{5}$$

Where: The decision trees are denoted by fk, the space of potential decision trees by F, and the number of boosting rounds by K. The model minimizes the following objective function:

$$L = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k) \tag{6}$$

Where: The loss function, represented by

$$l(y_i, \hat{y}_i) \tag{7}$$

, quantifies the variation between the observed and anticipated values. On the other hand, the regularization term, Ω(fk), regulates the intricacy of the trees, hence mitigating the risk of overfitting. XGBoost was evaluated using MAE, MAPE, and accuracy, and its performance was visualized by comparing actual vs. predicted prices.

6) LSTM : A recurrent neural network (RNN) version known as Long Short-Term Memory (LSTM) is specifically designed to capture the long-term reliance of sequential input. In order to control data flow and allow the network to store important patterns for extended periods of time, it is composed of memory cells and gates, namely input, output, and forget gates. Each LSTM cell contains three gates: The Forget Gate helps the Long Short-Term Memory (LSTM) to focus on the most crucial elements for future predictions by deciding which cell state data should be ignored.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{8}$$

Input Gate: Adds new data to the cells.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{9}$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{10}$$

Output Gate: This gate uses the cell state to determine the output.

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{11}$$

$$h_t = o_t \cdot \tanh(C_t) \tag{12}$$

Where: The input at time t is represented by the variable x_t, the hidden state from the previous time step by h_{t-1}, the cell state at time t by C_t, the sigmoid activation function by σ, which controls information flow by producing values between 0 and 1, and the hyperbolic tangent activation function by $\tanh$, which controls updates to the cell state by producing values between -1 and 1.

- **Performance Evaluation:** LSTM addresses the vanishing gradient problem that RNNs often encounter when

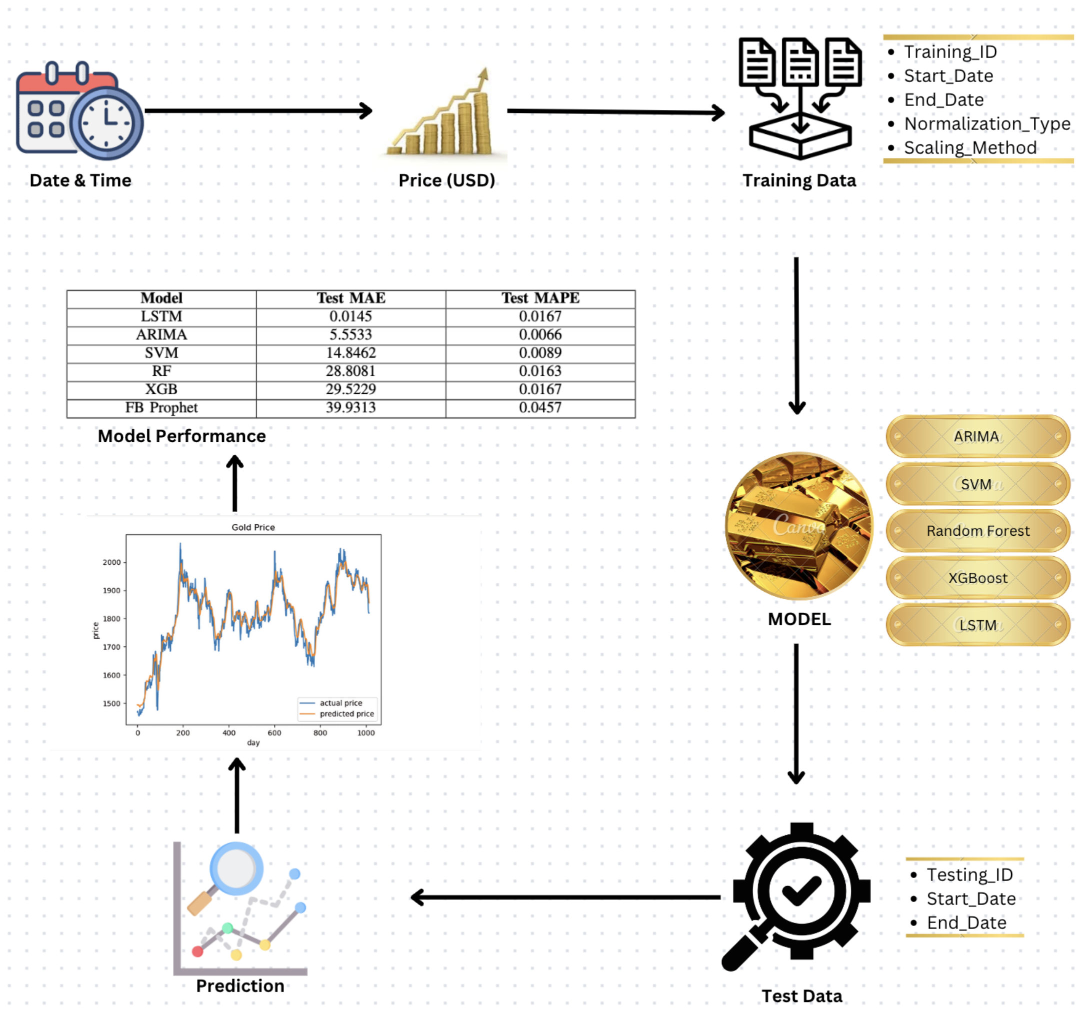

Fig. 1. Gold Price Model Configuration

capturing long-term dependencies. Its memory cell structure enables it to retain selectively or discard information with respect to time, making it well-suited for tasks like time-series forecasting. Particularly during times of market turbulence, LSTM has shown remarkable efficacy in learning both short-term oscillations and long-term patterns in the context of gold price prediction.The LSTM model was evaluated using MAE, MAPE, and accuracy. Its ability to capture sequential dependencies in gold prices made it the top-performing model in this study.

- **Metrics for Evaluation:** The performance of these models was evaluated and contrasted using the following metrics: The average magnitude of mistakes is determined using the Mean Absolute Error (MAE) measure. the differences in value between the expected and actual values, and it offers a simple way to gauge how accurate a forecast is. The model's accuracy is expressed as a percentage using the Mean Absolute Percentage Error (MAPE) metric, which provides a more comprehensible indicator of how well the predictions match the actual values in relation to each other.
- **Visualization and Comparison:** The performance of these models was evaluated and contrasted using the following metrics: The average magnitude of mistakes is determined using the Mean Absolute Error (MAE) measure. the differences in value between the expected and actual values, and it offers a simple way to gauge how accurate a forecast is. The model's accuracy is expressed as a percentage using the Mean Absolute Percentage Error (MAPE) metric, which provides a more comprehensible indicator of how well the predictions match the actual values in relation to each other.

IV. Result and Discussion

This study focused on forecasting gold prices using the United States Dollar (USD) as the standard currency. The various machine learning models applied include ARIMA, FB Prophet, Random Forest (RF), (SVM), XGBoost (XGB), and (LSTM). The results of Figure. 2 indicate that LSTM significantly outperformed other models in predicting gold prices over both the short and long term. In the domain of short-term projections, the ARIMA model, configured as (3, 1, 3) yielded good results, with a mean absolute error (MAE) of 5.5533 and a mean absolute percentage error (MAPE) of 0.0066. While it effectively captured linear relationships in gold price data, its limitations became evident in handling complex, nonlinear patterns, resulting in increasing prediction errors during periods of high volatility. In contrast, the LSTM model excelled with a MAE of 0.0145 and a MAPE of 0.0167, showcasing its ability to retain information over long sequences, which is important for accurately forecasting both immediate shifts and long-term patterns. The architecture of LSTM allowed it to learn sequential dependencies effectively, especially during volatile market periods. Although it showed a tendency to overfit on smaller datasets, this was mitigated through the use of dropout layers and careful hyperparameter tuning, confirming LSTM as a powerful tool for predicting gold prices.

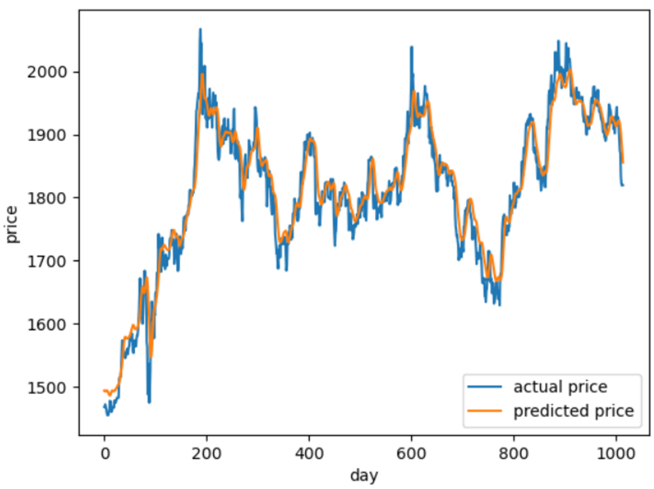

Fig. 2. Evaluating Real and Forecasted Gold Prices with the LSTM Model

FB Prophet performed reasonably well in capturing seasonal trends, achieving a MAE of 39.9313 and a MAPE of 0.0457. However, it struggled with sudden, nonlinear price changes, limiting its effectiveness during market volatility. Support Vector Machines (SVM) achieved a MAE of 14.8462 and a MAPE of 0.0089. While it effectively captured non-linear relationships, SVM required extensive tuning and was computationally intensive compared to simpler models. Random Forest demonstrated robust performance with a MAE of 28.8081 and a MAPE of 0.0163. It effectively handled nonlinear dependencies and reduced overfitting through ensemble learning, but its interpretability was limited compared to linear models. XGBoost provided the most accurate predictions among the traditional models, with a MAE of 29.5229 and a MAPE of 0.0167, showcasing its efficiency in managing large datasets and learning complex relationship Model

1) Discussion: The TABLE I findings of the study bring out the important role of machine learning technology such as LSTM in the financial time series forecasting, particularly

TABLE I
COMPARATIVE ANALYSIS OF MODEL PERFORMANCE USING TEST MAE AND TEST MAPE.

Model	Test MAE	Test MAPE
LSTM	0.0145	0.0167
ARIMA	5.5533	0.0066
SVM	14.8462	0.0089
RF	28.8081	0.0163
XGB	29.5229	0.0167
FB Prophet	39.9313	0.0457

gold prices. Because of its capacity to identify long-term dependencies and nonlinear relationships in the data, LSTM fared better than the other models in the research. Conversely yet, problem modeling occurs when it comes to the inability of the ARIMA and FB Prophet to handle the complex dynamics that govern long-term price dynamics. On the whole, the research outlines the primary characteristic of deep learning models like LSTM in time series forecasting. It is suggested that LSTM would be the optimal choice of the model to achieve the longest-term forecast accuracy regarding price movements of gold. This appearance could subsidize investors and politicians in making more informed decisions which rely on the predictions of the expected gold prices.

TABLE II
COMPARATIVE ANALYSIS OF MODEL PERFORMANCE USING TRAINING TIME AND INFERENCE TIME.

Model	Training Time	Inference Time
ARIMA	251.714 sec	0.55 sec
LSTM	96.16 sec	0.35 sec
FB Prophet	6.34 sec	1.45 sec
RF	2.90 sec	0.07 sec
XGB	0.65 sec	0.08 sec
SVM	0.62 sec	0.02 sec

The TABLE II elucidate distinct trade-offs between computational efficiency and model complexity. ARIMA and LSTM models exhibit higher computational demands during training, with LSTM compensating for this through relatively expeditious inference time, rendering it appropriate for complex, nonlinear data patterns. Conversely, models such as SVM and XGBoost demonstrate exceptional efficiency in both training and inference, making them optimal for real-time applications. FB Prophet, while exhibiting rapid training, is hindered by slower inference, thereby limiting its utility in dynamic environments. Tree-based models, including RF and XGBoost, offer a favorable balance of speed and versatility but may not capture intricate dependencies as effectively as LSTM. This analysis underscores the necessity of selecting models based on specific task requirements.

V. CONCLUSION & FUTURE WORK

In this study, we compared the performance of numerous machine learning algorithms, including ARIMA, LSTM, FB Prophet, SVM, Random Forest, and XGBoost, in predicting gold prices in the United States Dollar. To inventory the models, the LSTM had the highest performance, with the lowest mean absolute error (MAE) of 0.0145 and mean absolute percentage error (MAPE) of 0.0167. This might be owing to LSTM's ability to examine long-term correlations and accurately simulate irregular, non-linear relationships in time series data. While traditional models like ARIMA and FB Prophet performed well in capturing short-term trends and seasonal patterns, they struggled with the volatility and complexities of long-term price movements. The ensemble methods, particularly Random Forest and XGBoost, showed strong capabilities in handling nonlinear dependencies, though they were less accurate than LSTM. In future work, we aim to refine hyperparameter tuning for models like LSTM to further improve accuracy. Exploring hybrid models and incorporating external factors such as geopolitical events or market sentiment could enhance predictive capabilities, offering more robust tools for financial forecasting.

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